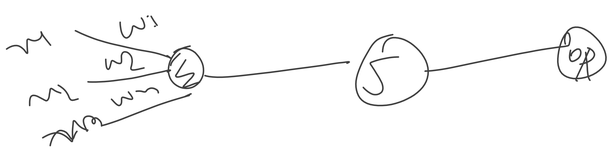
[redacted]

Feature	Perceptron Learning Rule	Gradient Descent
Data requirement	Requires data to be **linearly separable** to converge	Can handle **overlapping/non-linearly separable data**
Update method	Discrete updates: $w \leftarrow w + \eta(y - \hat{y})x$	Smooth updates using **gradient of a loss function**: $w \leftarrow w - \eta \frac{\partial L}{\partial w}$
Convergence guarantee	Only guaranteed if data is linearly separable	Can converge to **minimum of loss**, even if perfect separation not possible
Step control	Fixed learning rate; can overshoot	Learning rate can be tuned; can use **momentum**, **scheduling** to improve convergence
Error measurement	Implicit; only cares about correct/incorrect classification	Explicit; minimizes **continuous loss function** like MSE or cross-entropy
Flexibility	Works only for single-layer perceptron	Works for **single-layer and multi-layer networks** (MLPs)



$$L = \frac{1}{n} \sum_{i=1}^{n} \max(0, -y_i f(x_i))$$

x_1	x_2	y
x_{11}	x_{12}	y_1
x_{21}	x_{22}	y_2

$y_i = \{+1, -1\}$

$f(x_i) = f(w_1, w_2) \quad \boxed{w_1x_1 + w_2x_2}$

$$\max(0, -y_i f(x_i))$$

$x > 0 \quad x$

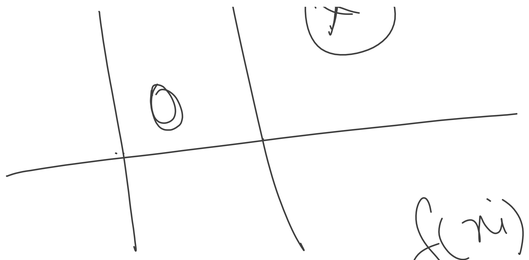
$x < 0 \quad 0$

Study hours	Attendance	y
2	3	+1
5	6	+1
1	2	-1
2	1	-1

y	$\hat{y}$	$\max(0, -y_i f(x_i))$
1	1	0
-1	-1	0
1	-1	
-1	1	

$\max\left(0, \underbrace{-y_i f(x_i)}_{\text{hinge}}\right) \quad (+1)\,x(+1)$

Cont'd



$f(x_i) \geq 0$

$f(x_i) < 0$

	x_1	x_2
1	x_{11}	x_{12}
2	x_{21}	

x_i x_{i1}

$$L = \underset{w_1, w_2, b}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} \max(0, -y_i f(x_i))$$

$$f(x_i) = w_1 x_{i1} + w_2 x_{i2} + b$$

$$w_1 = w_1 + \eta \frac{\partial L}{\partial w_1}, \quad w_2 = w_2 + \eta \frac{\partial L}{\partial w_2}, \quad b = b + \eta \frac{\partial L}{\partial b}$$

$$\left[\frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2}, \frac{\partial L}{\partial b}\right]$$

$$L = \frac{1}{n} \sum_{i=1}^{n} \max(0, -y_i f(x_i))$$

$$f(x_i) = w_1 x_{i1} + w_2 x_{i2} + b$$

$$\# \quad \frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial f(x_i)} \times \frac{\partial f(x_i)}{\partial w_1}$$

$$\frac{\partial L}{\partial f(x_i)} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y_i & \text{if } y_i f(x_i) < 0 \end{cases}$$

$$\frac{\partial f(x_i)}{\partial w_1} = x_{i1}$$

merge them together

$$\frac{\partial L}{\partial w_1} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y_i x_{i1} & \text{if } y_i f(x_i) < 0 \end{cases}$$

$$\# \ \frac{\partial L}{\partial w_2} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y_i x_{i2} & \text{if } y_i f(x_i) < 0 \end{cases}$$

$$\# \ \frac{\partial L}{\partial b} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y_i & \text{if } y_i f(x_i) < 0 \end{cases}$$